Nischal Bhandari

nischalbhandari73@gmail.com · (512) 779-2564 · Austin, TX
github.com/zkyko · linkedin.com/in/nischal-bhandari · zkyko.github.io/portfolio

Full-Stack Software Engineer | System Design · Data Modeling · Python · TypeScript · React

SUMMARY

Full-stack engineer who designs and architects systems end-to-end — with correctness verified, not assumed. 1.5 years at Four Hands shipping production platforms and AI systems (Python, TypeScript, React, PostgreSQL) with deterministic gates, security boundaries, and real failure diagnosis.

EXPERIENCE

Four Hands — Austin, TX Jul 2025 – Present
Software Engineer (Platform & Internal Tooling)

- **FH Connect Hub — system architecture:** designed and built the internal EDI platform that unified five systems (mailbox, Jira, Confluence, Mentis customer data, AI) into one app — **live in production**, with Logic App ingest → Azure SQL → queued workers → human review → ticket → knowledge note, Entra ID auth, scoped Key Vault secrets.
- **Three AI systems in production — system design:** test generation (ticket → Playwright spec), automated PR review, and request-to-PR delivery. Deterministic gates (tag → content-sufficiency → human approval) before any generation; always opens a PR so generated code is never lost; failure diagnosis reads real execution logs to separate test-script bugs from app bugs — *verified: caught a wrong login-flow assumption, not an outage*.
- **fh-d365-mcp — security architecture:** read-only MCP gateway for Dynamics 365; a policy layer rejects non-approved hosts/writes before a token is attached; TLS always on; zero stored secrets; read-only enforced by design.
- **Data modeling & reliability:** normalized domain schema (client → account → position → order → execution), migrations, JWT ownership checks; own CI/CD + observability — **500+ E2E tests**; **root-caused 5+ silent production failures** (incl. a stray `$schema` backslash that silently blocked redeploys for days, and a URL-concat bug caught via *getaddrinfo*).

ZT Systems (acquired by AMD) — Georgetown, TX Jun 2024 – Sep 2024
System Test Engineer / QA Analyst

- Automated RAID/BIOS server validation in Python + Bash; root-cause failure analysis on 75+ hardware configuration issues.

PROJECTS

StockTradingApp — *Python, FastAPI, SQLAlchemy, PostgreSQL*
Domain-first data model: normalized relational schema with table-per-subtype orders, Alembic migrations, and ownership-enforced APIs.

QA Studio / RecorderApp — *TypeScript, React, Electron, Playwright*
Workspace-based plugin architecture: recorder → locator registry → codegen pipeline, deterministic output for reviewable diffs.

Account Insights — *Salesforce LWC, Apex, GraphQL*
Snapshot worker built from chained, checkpointed jobs with run tokens and per-page data-accuracy rules; scores reconciled exact against production.

d365fo-client — *Python, MCP*
Read-only-by-default MCP server with a request policy gate in front of every call.

SKILLS

Design & Architecture: system design, data modeling, API design, security (least-privilege), deterministic verification
Languages: Python, TypeScript, JavaScript, SQL, Apex
Frontend / Backend: React, Angular, Next.js, LWC · FastAPI, Node.js, Express, GraphQL
Data: PostgreSQL, SQLAlchemy, Azure SQL, Alembic
Cloud, CI/CD & Testing: Docker, Linux, Azure, GitHub Actions, Azure DevOps, Playwright, BrowserStack

EDUCATION

B.S. Computer Science *(Minor in Mathematics)* — Texas State University Graduated 2025